

Danger Zone Analytics & Spatial Proximate Alerts Module Test Cases

- DNG-001 Alert Within 25km
- DNG-002 Outside 25km
- DNG-003 Multiple Citizens
- DNG-004 Multiple Incidents
- DNG-005 Alert Removed
- DNG-006 Critical Alert
- DNG-007 Medium Alert
- DNG-008 Safe Area

Field	Output Details
Test ID	DNG-001
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	In-Zone Proximity Danger Trigger (< 25km)
Actual Result	Incident 6a7806ea881fb90993f534f7 created at active coordinates. System established hazard radius, auto-assigned authority, and dispatched proximity notification triggers in 1034.08 ms .
Status	PASS

POST http://localhost:5000/api/

POST http://localhost:5000/api/

+

...

http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

JSON

Beautify

1

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7

"type": "Flood",

"description": "Rapid water level increase near main bridge",

"latitude": 6.8402,

"longitude": 79.9998,

"severity": "High"

Body

Cookies

Headers (8)

Test Results

Status: 201 Created

Time: 1100 ms

Size: 863 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

19

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"cluster_id": "6a7806ea881fb90993f534f6",

"verified_by": [],

"reported_inaccurate_by": [],

"_id": "6a7806ea881fb90993f534f7",

"status_history": [

{

"status": "Assigned",

"timestamp": "2026-08-09T04:49:47.362Z",

"changed_by": null,

"_id": "6a7806eb881fb90993f534fa"

}

],

"timestamp": "2026-08-09T04:49:46.940Z",

"_v": 1

Incoming API Request: [POST] /api/incidents

Time: 10:19:46 AM

```
-----  
BODY: {  
  type: 'Flood',  
  description: 'Rapid water level increase near main bridge',  
  latitude: 6.8402,  
  longitude: 79.9998,  
  severity: 'High'  
}
```

SEVERITY: High

🔍 Searching cluster for: 6.8402 79.9998

📍 Nearby incidents found: 0

❌ No cluster found. Creating new cluster (CLS-006)

```
NEW INCIDENT: {  
  user_id: new ObjectId('69f03beca1fb4905833b900e'),  
  type: 'Flood',  
  description: 'Rapid water level increase near main bridge',  
  image: null,  
  location: { type: 'Point', coordinates: [ 79.9998, 6.8402 ] },  
  assignedAuthorities: [],  
  severity: 'High',  
  status: 'Pending',  
  assigned_at: null,  
  cluster_id: new ObjectId('6a7806ea881fb90993f534f6'),  
  verified_by: [],  
  reported_inaccurate_by: [],  
  _id: new ObjectId('6a7806ea881fb90993f534f7'),  
  status_history: [],  
  timestamp: 2026-08-09T04:49:46.940Z  
}
```

```
SAVED INCIDENT: {  
  user_id: new ObjectId('69f03beca1fb4905833b900e'),  
  type: 'Flood',  
  description: 'Rapid water level increase near main bridge',  
  image: null,  
  location: { type: 'Point', coordinates: [ 79.9998, 6.8402 ] },  
  assignedAuthorities: [],  
  severity: 'High',  
  status: 'Pending',  
  assigned_at: null,  
  cluster_id: new ObjectId('6a7806ea881fb90993f534f6'),  
  verified_by: [],  
  reported_inaccurate_by: [],  
  _id: new ObjectId('6a7806ea881fb90993f534f7'),  
  status_history: [],  
  timestamp: 2026-08-09T04:49:46.940Z,  
  __v: 0  
}
```

```
  assigned_at: null,  
  cluster_id: new ObjectId('6a7806ea881fb90993f534f6'),  
  verified_by: [],  
  reported_inaccurate_by: [],  
  _id: new ObjectId('6a7806ea881fb90993f534f7'),  
  status_history: [],  
  timestamp: 2026-08-09T04:49:46.940Z,  
  __v: 0  
}
```

👤 Auto-assigning responder "Fire Officer Nimal" (69c8f887ac2b0d97336825ce) to incident "6a7806ea881fb90993f534f7"

✅ Assignment notifications created

✅ Status change notification created

POST /api/incidents 201 589 - 1034.084 ms

📄

Field	Output Details
Test ID	DNG-002
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Out-of-Range Proximity Filtering (> 25km)
Actual Result	Incident 6a780873881fb90993f535a3 created at distant coordinates ([80.6337, 7.2906]). Spatial geofencing prevented alert notifications from dispatching to users outside the active hazard radius.
Status	PASS

POST http://localhost:5000/api/incidents

POST http://localhost:5000/api/incidents

Save

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

JSON

Beautify

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8

"type": "Landslide",

"description": "Out of range proximity test incident",

"latitude": 7.2906,

"longitude": 80.6337,

"severity": "Moderate"

Body

Cookies

Headers (8)

Test Results

Status: 201 Created

Time: 2.55 s

Size: 864 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

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"cluster_id": "6a780873881fb90993f535a2",

"verified_by": [],

"reported_inaccurate_by": [],

"_id": "6a780873881fb90993f535a3",

"status_history": [

{

"status": "Assigned",

"timestamp": "2026-08-09T04:56:20.561Z",

"changed_by": null,

"_id": "6a780874881fb90993f535a6"

}

],

"timestamp": "2026-08-09T04:56:19.499Z",

"__v": 1

📡 Incoming API Request: [POST] /api/incidents

🕒 Time: 10:26:18 AM

BODY: {
 type: 'Landslide',
 description: 'Out of range proximity test incident',
 latitude: 7.2906,
 longitude: 80.6337,
 severity: 'Moderate'
}

SEVERITY: Moderate

🔍 Searching cluster for: 7.2906 80.6337

📍 Nearby incidents found: 0

❌ No cluster found. Creating new cluster (CLS-006)

NEW INCIDENT: {
 user_id: new ObjectId('69f03beca1fb4905833b900e'),
 type: 'Landslide',
 description: 'Out of range proximity test incident',
 image: null,
 location: { type: 'Point', coordinates: [80.6337, 7.2906] },
 assignedAuthorities: [],
 severity: 'Moderate',
 status: 'Pending',
 assigned_at: null,
 cluster_id: new ObjectId('6a780873881fb90993f535a2'),
 verified_by: [],
 reported_inaccurate_by: [],
 _id: new ObjectId('6a780873881fb90993f535a3'),
 status_history: [],
 timestamp: 2026-08-09T04:56:19.499Z
}

SAVED INCIDENT: {
 user_id: new ObjectId('69f03beca1fb4905833b900e'),
 type: 'Landslide',
 description: 'Out of range proximity test incident',
 image: null,
 location: { type: 'Point', coordinates: [80.6337, 7.2906] },
 assignedAuthorities: [],
 severity: 'Moderate',
 status: 'Pending',
 assigned_at: null,
 cluster_id: new ObjectId('6a780873881fb90993f535a2'),
 verified_by: [],
 reported_inaccurate_by: [],
 _id: new ObjectId('6a780873881fb90993f535a3'),
 status_history: [],
 timestamp: 2026-08-09T04:56:19.499Z,
 __v: 0
}

👤 Auto-assigning responder "Jane Pending Responder" (6a27003ff2e6e5644469301d) to incident "6a780873881fb90993f535a3"

✅ Assignment notifications created

✅ Status change notification created

POST /api/incidents 201 590 - 2543.716 ms

📄

Field	Output Details
Test ID	DNG-003
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Spatial Broadcast to Multiple Citizens in Danger Radius
Actual Result	Authenticated secondary citizen account in Maharagama ([79.9233, 6.8481]) queried active hazards and received the Homagama incident (6a7806ea881fb90993f534f7) located 7.8km away in 368.23 ms .
Status	PASS

Field	Output Details
Test ID	DNG-004
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Multiple Active Incidents Overlay & Aggregation
Actual Result	Query successfully returned multiple concurrent incidents (6a7806ea881fb90993f534f7, 6a775307e265edbc5b7efd25, 6a77032554da6498db0eed31) across nearby coordinates, maintaining distinct cluster IDs (CLS-006, etc.) and severity levels.
Status	PASS

POST http://localhost:5000/ap POST http://localhost:5000/ap POST http://localhost:5000/ap GET http://localhost:5000/api + ...

http://localhost:5000/api/incidents Save

GET http://localhost:5000/api/incidents Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary

This request does not have a body

Body Cookies Headers (8) Test Results Status: 200 OK Time: 381 ms Size: 6.33 KB Save Response

Pretty Raw Preview Visualize JSON

```
318   "cluster_id": "6a76f29c46cc964d15d08a6c",
319   "verified_by": [],
320   "reported_inaccurate_by": [],
321   "status_history": [
322     {
323       "status": "Assigned",
324       "timestamp": "2026-08-08T09:10:52.852Z",
325       "changed_by": null,
326       "_id": "6a76f29c46cc964d15d08a70"
327     },
328   ],
329   "timestamp": "2026-08-08T09:10:52.620Z",
330   "__v": 1
331 }
332 ]
```

```
-----
Incoming API Request: [GET] /api/incidents
Time: 10:45:22 AM
-----

GET /api/incidents 200 6213 - 368.232 ms
```

Field	Output Details
Test ID	DNG-005
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Alert Removal & Spatial Hazard Resolution
Actual Result	Incident 6a780873881fb90993f535a3 updated to Resolved. Status change notification generated and active spatial danger warnings revoked in 528.04 ms .
Status	PASS

Incident Management

Active Incidents

Verified events requiring dispatch

Fire

Dispatched

Hiniduma-Opatha-Pitabeddara Road, Jesmin Valley Junction, Thawalama, Galle District, Southern Province, 81478, Sri Lanka

Aug 9, 2026, 10:51 AM

1 Units Assigned

Flood

Resolved

Homagama Bus Stand, Godaparagahalanda
* Mawatha, Homagama South, Homagama, Colombo District, Western Province, 10200, Sri Lanka

Aug 9, 2026, 10:19 AM

1 Units Assigned

Accident

Dispatched

Homagama Bus Stand, Godaparagahalanda
* Mawatha, Homagama South, Homagama, Colombo

Disaster

Resolved

Hiniduma-Opatha-Pitabeddara Road, Jesmin Valley Junction, Thawalama Junction, Thawalama, Galle District, Southern Province, 81478, Sri Lanka

ID: 6a78166cea1a68c3883ad6d6

Dispatch Responders

Auto-Dispatch ON

Live Activity Log

CURRENTLY ASSIGNED

ABD.pvt.ltd (D Silva)

EN ROUTE

AVAILABLE RESPONDERS (BY PROXIMITY)

Colombo East Police Devison Unit (Amal Perera)

+ Assign

0 m away

System Auto-Verified

10m ago

Reached citizen verification threshold.

Incident Registered

12m ago

Incident #1043 logged in database.

Dispatch Sent

Just now

1 units confirmed reception.

```
-----
Incoming API Request: [PUT] /api/incidents/6a780873881fb90993f535a3/status
Time: 11:48:40 AM
-----
[✓] Status change notification created
PUT /api/incidents/6a780873881fb90993f535a3/status 200 773 - 528.037 ms
```

Field	Output Details
Test ID	DNG-006
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Critical Danger Alert Trigger (severity: High)
Actual Result	Incident 6a781e435a0be73ee480a974 processed with priority tiering. Auto-assigned fire authority and initialized priority alert notifications in 6842.05 ms .
Status	PASS

POST http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

Beautify

none

form-data

x-www-form-urlencoded

raw

binary

JSON

1

2

3

4

5

6

7

Body

Cookies

Headers (8)

Test Results

Status: 201 Created

Time: 6.85 s

Size: 874 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

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```
PS G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend> node server.js
NEW INCIDENT: {
  user_id: new ObjectId('69f03beca1fb4905833b900e'),
  type: 'Fire',
  description: 'High-severity industrial chemical fire spreading rapidly',
  image: null,
  location: { type: 'Point', coordinates: [ 79.998, 6.8415 ] },
  assignedAuthorities: [],
  severity: 'High',
  status: 'Pending',
  assigned_at: null,
  cluster_id: new ObjectId('6a781e435a0be73ee480a973'),
  verified_by: [],
  reported_inaccurate_by: [],
  _id: new ObjectId('6a781e435a0be73ee480a974'),
  status_history: [],
  timestamp: 2026-08-09T06:29:23.983Z
}
SAVED INCIDENT: {
  user_id: new ObjectId('69f03beca1fb4905833b900e'),
  type: 'Fire',
  description: 'High-severity industrial chemical fire spreading rapidly',
  image: null,
  location: { type: 'Point', coordinates: [ 79.998, 6.8415 ] },
  assignedAuthorities: [],
  severity: 'High',
  status: 'Pending',
  assigned_at: null,
  cluster_id: new ObjectId('6a781e435a0be73ee480a973'),
  verified_by: [],
  reported_inaccurate_by: [],
  _id: new ObjectId('6a781e435a0be73ee480a974'),
  status_history: [],
  timestamp: 2026-08-09T06:29:23.983Z,
  __v: 0
}
🔴 Auto-assigning responder "Fire Officer Nimal" (69c8f887ac2b0d97336825ce) to incident "6a781e435a0be73ee480a974"
✅ Assignment notifications created
✅ Status change notification created
POST /api/incidents 201 600 - 6842.054 ms

```

Field	Output Details
Test ID	DNG-007
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Medium Danger Alert & Proximity Cluster Merging (severity: Moderate)
Actual Result	Incident 6a781f255a0be73ee480a97f processed in 527.34 ms . System detected active neighboring hazard node, dynamically attached incident to cluster 6a781e435a0be73ee480a973, and assigned responder.
Status	PASS

POST http://localhost:5000/api/

POST http://localhost:5000/api/

POST http://localhost:5000/api/

POST http://localhost:5000/api/

POST http://localhost:5000/api/

+ ...

http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

JSON

Beautify

1

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3

4

5

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7

"type": "Road Hazard",

"description": "Moderate fallen tree blocking single lane",

"latitude": 6.8420,

"longitude": 79.9970,

"severity": "Moderate"

Body

Cookies

Headers (8)

Test Results

Status: 201 Created

Time: 536 ms

Size: 869 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

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"assigned_at": null,

"cluster_id": "6a781e435a0be73ee480a973",

"verified_by": [],

"reported_inaccurate_by": [],

"_id": "6a781f255a0be73ee480a97f",

"status_history": [

{

"status": "Assigned",

"timestamp": "2026-08-09T06:33:09.680Z",

"changed_by": null,

"_id": "6a781f255a0be73ee480a982"

}

],

"timestamp": "2026-08-09T06:33:09.491Z",

```
PS G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend> node server.js
```

```
NEW INCIDENT: {
  user_id: new ObjectId('69f03beca1fb4905833b900e'),
  type: 'Road Hazard',
  description: 'Moderate fallen tree blocking single lane',
  image: null,
  location: { type: 'Point', coordinates: [ 79.997, 6.842 ] },
  assignedAuthorities: [],
  severity: 'Moderate',
  status: 'Pending',
  assigned_at: null,
  cluster_id: new ObjectId('6a781e435a0be73ee480a973'),
  verified_by: [],
  reported_inaccurate_by: [],
  _id: new ObjectId('6a781f255a0be73ee480a97f'),
  status_history: [],
  timestamp: 2026-08-09T06:33:09.491Z
}
```

```
SAVED INCIDENT: {
  user_id: new ObjectId('69f03beca1fb4905833b900e'),
  type: 'Road Hazard',
  description: 'Moderate fallen tree blocking single lane',
  image: null,
  location: { type: 'Point', coordinates: [ 79.997, 6.842 ] },
  assignedAuthorities: [],
  severity: 'Moderate',
  status: 'Pending',
  assigned_at: null,
  cluster_id: new ObjectId('6a781e435a0be73ee480a973'),
  verified_by: [],
  reported_inaccurate_by: [],
  _id: new ObjectId('6a781f255a0be73ee480a97f'),
  status_history: [],
  timestamp: 2026-08-09T06:33:09.491Z,
  __v: 0
}
```

```
🔴 Auto-assigning responder "Fire Officer Nimal" (69c8f887ac2b0d97336825ce) to incident "6a781f255a0be73ee480a97f"
```

```
✅ Assignment notifications created
```

```
✅ Status change notification created
```

```
POST /api/incidents 201 595 - 527.344 ms
```

Field	Output Details
Test ID	DNG-008
Module	Danger Zone Analytics & Spatial Proximity Alerts
Feature	Safe Area Boundary & Distance Filter Transition
Actual Result	Updating citizen location to Galle ([80.221, 6.0535]) successfully transitioned active danger status to Safe in 1080.65 ms , confirming spatial geofence clearance beyond 25km.
Status	PASS

```
_id: ObjectId('6a23cdc4b9b78da402d0f380')
name: "N.D. Ashen Perera"
email: "ashen.p01@gmail.com"
password: "$2b$10$mNZUZ5Czv/0mWVIDaAflqufogmYjue3/R2gR8QhkE//rzGiVG8tha"
role: "Citizen"
district: "Colombo"
▼ location: Object (2)
  type: "Point"
  ▼ coordinates: Array (2)
    0: 80.221
    1: 6.0535
organization: "-"
contact_number: "0712345601"
isVerified: true
status: "Active"
registered_date: 2026-06-06T07:35:32.041+00:00
__v: 0
```

POST http://localhost:5000/api/ POST http://localhost:5000/api/ POST http://localhost:5000/api/ POST http://localhost:5000/api/ GET http://localhost:5000/api/ + ...

http://localhost:5000/api/incidents Save

GET http://localhost:5000/api/incidents Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Type Bearer Token Token eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ...

The authorization header will be automatically generated when you send the request. [Learn more about authorization](#)

Store your secrets with end-to-end encryption locally using Vault. Store in Vault

Body Cookies Headers (8) Test Results Status: 200 OK Time: 1087 ms Size: 7 KB Save Response

Pretty Raw Preview Visualize JSON

```
359   "cluster_id": "6a77f50ceaff9af045775648",
360   "verified_by": [],
361   "reported_inaccurate_by": [],
362   "status_history": [
363     {
364       "status": "Assigned",
365       "timestamp": "2026-08-09T03:33:33.120Z",
366       "changed_by": null,
367       "_id": "6a77f50deaff9af04577564c"
368     }
369   ],
370   "timestamp": "2026-08-09T03:33:32.916Z",
371   "__v": 1
372
373 ]
```

```
-----
Incoming API Request: [GET] /api/incidents
Time: 12:19:12 PM
-----

GET /api/incidents 200 6899 - 1080.649 ms
```